MICHAEL P. RAGSDALE

Senior Frontend Systems Engineer | Design Systems Architect Design Systems • UI Component Libraries • Web Accessibility (WCAG) • Tailwind CSS Norfolk, VA | U.S. Citizen | hireme@michaelpragsdale.com

Professional Summary

Senior Frontend Systems Engineer with over a decade of experience designing, architecting, and evolving production-grade design systems and reusable component libraries. Passionate about bridging the gap between product design and engineering by building robust, accessible, and ergonomic frontend foundations for data-dense workflows. Expert in modern frontend architecture, Tailwind CSS, semantic HTML primitives, and rigorous accessibility standards (WCAG 2.1 AA). Proven ability to zoom in on the micro-interactions of a single form control and zoom out to shape the macro-architecture of an enterprise platform.

Technical Skills

- **Frontend Systems:** Component Library Architecture, SPA Frameworks (React concepts / Custom Vanilla JS rendering), Tailwind CSS, SCSS, HTML5 Primitives, Responsive Design
- **Accessibility (a11y):** WCAG 2.1 AA, Section 508, Keyboard Navigation, Screen Reader Support (NVDA/JAWS), High-Contrast/Reduced-Motion paradigms
- **Design & Workflow:** Figma, UI/UX Prototyping, Data-Dense Dashboards, Design-to-Code Translation, Technical Documentation
- **Infrastructure:** JavaScript (ES6+), TypeScript fundamentals, JSON Routing, Git, CI/CD, Agile Methodologies

Relevant Technical Experience

Principal Systems Architect & Founder

RaggieSoft Media | Norfolk, VA | raggiesoftmedia.com | Est. 2008 – Present Founded and operate RaggieSoft Media, an independent software development and digital media publishing ecosystem supporting web applications, cloud infrastructure, narrative publishing, music distribution, and open-source platform development.

- **Design System Architecture:** Architected and open-sourced **Stardust Engine CMS**, an MIT-licensed, zero-database PHP MVC ecosystem powering 9 production domains. Built a unified, reusable component library to ensure visual consistency and developer ergonomics across the entire platform.
- **Component Engineering & Tailwind CSS:** Leading the frontend architectural design for an upcoming media distribution and music player platform. Utilizing **Tailwind CSS** to build composable, utility-driven UI primitives and higher-order application blocks that scale across complex, data-rich media workflows.
- **Multi-Domain & Cloud Edge:** Orchestrate a production fleet of 9 distinct domains via **Cloudflare DNS/CDN** edge routing, 301 redirection rules, and automated deployment pipelines syncing assets to **DigitalOcean Spaces** (S3-compatible object storage) and Nginx reverse proxies.
- **Automated Media Engineering:** Engineered automated server-side pipelines utilizing **FFmpeg** and Bash to transcode and normalize audio assets across multiple distribution formats, paired with custom **PHP GD** engines that dynamically composite Open Graph social cards across routes.
- **E-Commerce & Digital Fulfillment:** Integrated **Fourthwall** as an international Merchant of Record (MoR) with custom DNS architecture, powering automated fulfillment of physical merchandise and digital assets without third-party vendor lock-in.
- **Accessibility by Design:** Enforce strict **WCAG 2.1 AA** and **Section 508** compliance across all UI components, custom modals, media players, and narrative readers, ensuring zero accessibility debt out of the box.

Accessible Document Engineer (Data Workflows)

City of Virginia Beach Parks & Recreation | Virginia Beach, VA | May 2023 – Present

- **Data-Dense Interface Translation:** Architect manual ETL workflows to extract complex, data-heavy operational reporting from legacy enterprise platforms (ActiveNet) and translate them into clean, highly readable, and accessible digital structures.
- **Workflow Optimization:** Analyze real-world administrative workflows (triage, configuration, reporting) to design system processes that improve usability and data accuracy for municipal staff.

Web Accessibility Specialist (Computer Operations)

Old Dominion University (ODU Online) | Norfolk, VA | Nov 2019 – Jul 2022

- **Pattern Library & Templates:** Engineered a universal, inherently accessible pattern library and page-level templates deployed across the university's Canvas LMS ecosystem, ensuring cross-platform consistency for thousands of users.
- **Inclusive Experience Design:** Executed rigorous manual testing and remediation of complex frontend codebases, guaranteeing seamless keyboard navigation and screen-reader support (NVDA/JAWS) for disabled students and faculty.
- **Cross-Functional Mentorship:** Collaborated closely with instructional designers, product managers, and developers to advocate for inclusive design principles, proving that strict accessibility compliance does not compromise visual elegance.

Education & Certification

- **Tidewater Community College | Virginia Beach, VA** Associate of Science, Information Technology (Software Engineering) | Expected Dec 2027
- **Old Dominion University | Norfolk, VA** Bachelor of Science, Customized Studies in Leadership | Prior Coursework
- **Wilson Workforce and Rehabilitation Center | Fishersville, VA** Certificate, Business & Information Technology | 2011